Gravity as Relational Difference Elimination: A Harmonic Closure Derivation

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Model status. This manuscript is an exploratory, falsifiable framework. Claims are labeled throughout using the taxonomy in Table 1. No experimental validation independent of the author's prior program is claimed. The coupling α is calibrated (one free parameter). A relativistic completion does not yet exist.

Claim Type	Description
[POSTULATE]	A foundational assumption — not derived, not yet falsified. A derivation from more primitive axioms is an open question.
[DERIVED]	A mathematical consequence of the postulates. Can be checked for internal consistency.
[CALIBRATION]	A parameter fixed by matching to an empirical value. Shifts predictive power to variations, not absolute values.
[HYPOTHESIS]	A testable physical conjecture requiring external validation.
[SPECULATIVE]	An exploratory program note. Not yet a genuine prediction.

Table 1: Claim-type legend used throughout the manuscript.

Abstract

We propose a relational–informational model in which gravity emerges as the macroscopic tendency of physical systems to eliminate differences between projected relational states. The framework is based on a universal harmonic partition of system frequencies $\omega_N^2 = \omega_V^2 + \omega_m^2$, previously applied within the same research program to hadronic, atomic, and cosmological systems. Within this structure the gravitational source is identified as the informational density of confinement $\rho_{\mathrm{rel}} = \kappa\omega_m^2\rho_m$. This leads to a Poisson equation $\nabla^2\phi = 4\pi\alpha\kappa\omega_m^2\rho_m$, recovering Newtonian gravity with $G = \alpha\kappa\omega_m^2$. In this interpretation Newton's constant is not fundamental but emerges from the interaction between confinement physics and relational projection geometry. The framework connects proton structure, valence chemistry, cosmic expansion, and gravitational interaction through a single harmonic closure principle spanning approximately 46 orders of magnitude. Falsifiable predictions include: composition-dependent deviations from the weak equivalence principle, corrections near the closure scale detectable in neutron-star observations, and scale-dependent effective gravitational coupling that may mimic dark-matter phenomenology under partial closure conditions. The coupling α is calibrated to one empirical input (CODATA 2018 value of G); all other quantities are fixed by prior program results.

Keywords: emergent gravity; relational physics; harmonic partition; binary closure; unification; weak equivalence principle

Originality and Novelty Statement

What is novel. This work derives Newton's constant G as an emergent quantity from a pre-existing relational-projection algebra, rather than treating it as a primitive input. Previous emergent-gravity approaches (Verlinde [4], Jacobson [3], Padmanabhan [5]) reproduce Newtonian gravity qualitatively from thermodynamic analogies but leave G as a proportionality constant. This model expresses G in terms of microscopically meaningful quantities — a confinement frequency ω_m, a closure-cell volume κ, and a universal dimensionless coupling α — thereby shifting predictive power to composition-dependent variations of G.

How it differs from existing literature. The relational projection framework unifies four distinct physical scales (hadronic, atomic, cosmological, gravitational) from a single algebraic postulate — $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure — without retrofitting each domain independently. The key distinction from thermodynamic approaches is structural: entropic and holographic frameworks assign temperature and entropy to macroscopic screens, but have no microscopic mechanism that determines the numerical value of G. The present framework assigns a specific geometric volume κ to each nucleon's closure state, making G explicitly dependent on hadronic-scale physics and therefore in principle sensitive to composition. This cross-scale quantitative closure is absent from all comparison frameworks listed in Table 5.

Acknowledged limitations. The coupling α is calibrated, not derived (one free parameter). The framework is valid only in the weak-field, stationary regime. A relativistic tensor completion remains open. The physical correspondence between the relational-charge ansatz and quantum field theory is not established. The selection rule fixing binary depth n for each system is not proven as a theorem — only structural motivation exists. Prior applications of the harmonic partition (proton radius, valence stability, Hubble expansion) are internal consistency results from the same research program and are not claimed as independent validations.

1 Introduction

Gravity remains the only fundamental interaction whose coupling constant G is imposed empirically rather than derived from deeper principles. While General Relativity provides an exquisite classical description, its conceptual origin — why mass curves spacetime with precisely this strength — remains unexplained. The question this paper addresses is: *can the value of G be expressed in terms of pre-existing microscopic quantities, rather than treated as a primitive input to the theory?*

Within the author's research program, a harmonic partition of system frequencies has been applied (not independently validated) to three domains, yielding quantitative results that motivate the present extension:

Scale	**System**	**Harmonic ratio**	**Result**
Hadronic	Proton charge radius	$\omega_V^2 : \omega_m^2 = 1{:}15$	$r_p = 4\hbar/(m_p c) \approx 0.8412$ fm
Atomic	Valence stability (s-p)	$a : b$ prime or 2^n	Harmonic weight $P = a + b$
Cosmological	Hubble expansion	$V = \omega R$	$H_0 = \omega_V$ when $R = R_H$

Table 2: Prior applications of the relational projection framework within this research program. These are internal consistency results, not independent validations.

This paper extends the same principle to gravity. We show that Newton's constant G is not fundamental but emerges as

$$G = \alpha \kappa \omega_m^2 \tag{1}$$

where ω_m is the confinement frequency (hadronic scale), κ is the specific relational volume of the harmonic closure state, and α is a universal dimensionless coupling. **[CALIBRATION]** The framework does not predict the absolute numerical value of G without calibration — it expresses G in terms of microscopically meaningful quantities, shifting predictive power to variations when composition or closure state changes.

2 Relational Foundations

2.1 The Harmonic Partition Postulate

[POSTULATE] 1 (Universal Harmonic Partition). Every stable relational system admits an orthogonal decomposition of its total frequency ω_N:

$$\omega_N^2 = \omega_V^2 + \omega_m^2 \tag{2}$$

where ω_V is the projection frequency (visible, relational mode), ω_m is the confinement frequency (inertial, informational mode), and $\omega_N = mc^2/\hbar$ for massive systems. **[POSTULATE]** This is treated as a physical constraint selecting stable configurations through rational ratios $\omega_V : \omega_m$. A derivation of Eq. (2) from more primitive axioms is an open question.

2.2 Binary Closure Principle

[POSTULATE] 2 (Binary Information Closure). A relational system achieves maximal stability when its harmonic partition satisfies a binary closure of depth n:

$$\frac{\omega_V^2}{\omega_N^2} = \frac{1}{2^n}, \qquad n \in \mathbb{N} \tag{3}$$

Observed ratios used in this program: $n = 2 \to \omega_V^2 : \omega_m^2 = 1{:}3$ (atomic valence); $n = 4 \to \omega_V^2 : \omega_m^2 = 1{:}15$ (hadronic). **[POSTULATE]** The selection rule fixing n for each physical system is an open problem. Structural motivation based on spin-$\frac{1}{2}$ and the real representation dimension of $\mathfrak{su}(2)$ is discussed in prior work, but a derivation theorem does not yet exist.

2.3 Identity Radius as Derived Quantity

[DERIVED] The visible radius is defined operationally as $R_V \equiv c/\omega_V$. For a nucleon in closure at depth $n = 4$: $\omega_V = \omega_p/4$, giving the nucleonic identity radius:

$$R_{\text{id}}^{(\text{nuc})} = R_V = 4\bar{\lambda}_p = \frac{4\hbar}{m_p c} \approx 0.8412 \text{ fm} \tag{4}$$

[HYPOTHESIS] This numerical proximity to the CODATA 2018 proton charge radius (0.8414 ± 0.0019 fm, [1]) and to the muonic hydrogen determination (0.84087 ± 0.00039 fm, [2]) is treated as a falsifiable consistency check of the $n = 4$ assignment, not as a confirmed independent prediction. For a composite body of N nucleons, the effective identity radius scales as $R_{\text{id}} = N^{1/3} R_{\text{id}}^{(\text{nuc})}$.

3 Gravitational Model: Derivation from Harmonic Closure

3.1 Assumptions (Weak-Field Closure Regime)

The following assumptions define the domain of validity of the present derivation (see also [8] for the standard treatment of the weak-field limit):

(A1) Stationary, weak-field limit: no retardation, no radiation.

(A2) Euclidean 3-space and spherical symmetry outside sources.

(A3) Conservative potential field: $\mathbf{g} = -\nabla\phi$.

(A4) Outside sources, $\nabla^2\phi = 0 \Rightarrow$ unique $1/r$ solution $\Rightarrow \mathbf{g} \propto 1/r^2$.

(A5) Matching to Newton fixes only the normalization of ϕ in the classical limit.

3.2 Gravitational Source Identification

[HYPOTHESIS] Within this framework, mass is tied to confinement through ω_m. The simplest scalar source compatible with Eq. (2) in the weak-field regime is taken as Source $\propto \omega_m^2 \rho_m$ — the informational content of confinement per unit volume. This choice is a *minimal ansatz*, not a derived result. Alternative couplings are possible and would alter the structure of the WEP predictions in Section 5.1.

3.3 Specific Relational Volume (Closure Cell)

[DERIVED] For nucleonic matter in harmonic closure at binary depth n, the closure-cell volume per nucleon is:

$$v_{\mathrm{id}} = \frac{4\pi}{3}\left(R_{\mathrm{id}}^{(\mathrm{nuc})}\right)^3 = \frac{4\pi}{3}\left(2^{n/2}\bar{\lambda}_p\right)^3 \tag{5}$$

[DERIVED] The specific relational volume (volume per unit mass) is then

$$\kappa \equiv \frac{V_{\mathrm{cl}}}{M} = \frac{v_{\mathrm{id}}}{m_p} \tag{6}$$

Crucially, κ depends on closure state (via n) and m_p, but not on system size N. This preserves $Q \propto M$ and Newtonian $1/r^2$ scaling.

3.4 Relational Charge and Poisson Equation

[HYPOTHESIS] Define the relational charge density

$$\rho_{\mathrm{rel}} \equiv \kappa\omega_m^2\rho_m \tag{7}$$

where ρ_m is ordinary mass density. Introducing a dimensionless coupling α, the gravitational potential satisfies:

$$\nabla^2\phi = 4\pi\alpha\rho_{\mathrm{rel}} = 4\pi\alpha\kappa\omega_m^2\rho_m \tag{8}$$

[DERIVED] Dimensional check: $[\nabla^2\phi] \sim [\mathrm{T}]^{-2}$, $[\rho_m] \sim [\mathrm{M}][\mathrm{L}]^{-3}$, $[\kappa] \sim [\mathrm{L}]^3[\mathrm{M}]^{-1}$, $[\omega_m^2] \sim [\mathrm{T}]^{-2}$, $[\alpha] \sim 1$: right-hand side $\sim [\mathrm{T}]^{-2}$. ✓

3.5 Newton's Constant as a Derived Quantity

[DERIVED] For a point mass M, Eq. (8) gives $\phi(r) = -\alpha\kappa\omega_m^2 M/r$. Comparing with Newton's $\phi_N(r) = -GM/r$ yields:

$$\boxed{G = \alpha\kappa\omega_m^2} \tag{9}$$

[CALIBRATION] This is not a prediction of the absolute value of G without calibration. For nucleonic matter in ground-state closure ($n = 4$): $\kappa \approx 1.49 \times 10^{-18}\ \mathrm{m^3\,kg^{-1}}$, $\omega_m^2 \approx 1.905 \times 10^{48}\ \mathrm{s^{-2}}$, giving $\kappa\omega_m^2 \approx 2.84 \times 10^{30}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$. Setting $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ (CODATA 2018 [1]) fixes:

$$\alpha \approx 2.35 \times 10^{-41} \tag{10}$$

3.6 Parameter Accounting: The Coupling α

4 Unification Across Scales

[DERIVED] All four physical domains follow from $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure:

Property	Description
Definition	Dimensionless universal coupling in the relational Poisson equation.
How fixed	**[CALIBRATION]** Set by matching $G = \alpha\kappa\omega_m^2$ to CODATA 2018 value of G. This is the model's only empirical input.
Numerical value	$\alpha \approx 2.35 \times 10^{-41}$
Sensitivity	If ω_m or κ vary with composition, α translates that variation into gravitational anomalies. Assumed universal — this assumption is itself falsifiable.
Open question	Deriving α from a higher-order harmonic closure condition is listed as explicit future work.

Table 3: Parameter accounting for the dimensionless coupling α.

Scale	System	n	ω_V	ω_m	Emergent property
Hadronic	Proton	4	$\omega_p/4$	$\sqrt{15}/4\,\omega_p$	Charge radius ≈ 0.8412 fm
Atomic	s-p valence	1,2	N_s freq.	N_p freq.	Harmonic weight P
Cosmological	Hubble flow	0 (lim.)	H_0	$\to 0$	Expansion law $V = H_0 D$
Gravitational	Newtonian	4 (nuc.)	$c/R_{\rm id}$	ω_m	$G = \alpha\kappa\omega_m^2$

Table 4: Quadripartite unification via $\omega_N^2 = \omega_V^2 + \omega_m^2$. Spans ~ 46 orders of magnitude from a single harmonic closure principle.

5 Testable Predictions and Falsifiability

The following are presented as genuine falsifiable targets. Each could, in principle, rule out the framework or force reinterpretation.

5.1 Composition Dependence — Weak Equivalence Principle

[HYPOTHESIS] If $\omega_{m,t}$ (or κ_t) varies slightly with composition — e.g., via binding energy per nucleon — the model predicts small WEP violations. The Eötvös parameter:

$$\eta_{A,B} \approx \frac{\Delta(\omega_m^2)}{\omega_m^2} + \frac{\Delta\kappa}{\kappa} \tag{11}$$

FALSIFICATION CONDITION: This model predicts η should correlate with *mean binding energy per nucleon*, not only with atomic mass. Current limits from MICROSCOPE [6] give $\eta \lesssim 10^{-15}$. The model is falsified (or severely constrained) if η shows no composition-binding-energy correlation at precisions achievable by next-generation equivalence-principle experiments (e.g., STE-QUEST, MICROSCOPE follow-on). A more quantitative estimate requires an explicit microphysical map from nuclear composition to ω_m within the program.

5.2 High-Density Corrections Near Closure Scale

[HYPOTHESIS] When $r \lesssim R_{\rm id}$, the point-source approximation fails. The potential acquires corrections:

$$\phi(r) = -\frac{GM}{r}\left[1 + \beta\left(\frac{R_{\rm id}}{r}\right)^2 + \gamma\left(\frac{R_{\rm id}}{r}\right)^4 + \cdots\right] \tag{12}$$

FALSIFICATION CONDITION: For neutron stars ($N \sim 10^{57}$), $R_{\rm id} \sim N^{1/3} R_{\rm id}^{(\rm nuc)} \sim 10^4$ m. If neutron-star merger observations (LIGO/Virgo tidal deformability) or pulsar-timing data show no deviation from Newtonian point-mass behavior at scales comparable to $R_{\rm id}$, this correction structure is ruled out. Note: the coefficients β, γ are not yet derived from the postulates — this prediction is qualitative until a relativistic completion is specified.

5.3 Scale-Dependent Gravity and Dark-Matter Phenomenology

[SPECULATIVE] In diffuse systems (e.g., galaxies), if closure is partial, the effective specific volume $\kappa(r)$ becomes scale-dependent, giving $G_{\rm eff}(r) = \alpha\kappa(r)\omega_m^2$. A minimal phenomenological ansatz $\kappa(r) = \kappa_0[1 + \lambda(r/r_0)^\delta]$ can produce flat rotation curves.

FALSIFICATION CONDITION: Rotation-curve anomalies should correlate with informational density $\omega_m^2\rho_m$, not only with total mass distribution. This prediction distinguishes the framework from both ΛCDM and MOND [9, 10]. *Important caveat: the ansatz for $\kappa(r)$ is not derived from the postulates. This is a program note, not a genuine prediction. It is listed here for transparency and to invite critique, not to claim explanatory priority over MOND or ΛCDM.*

6 Discussion

6.1 Comparison with Other Emergent Gravity Frameworks

Approach	**Source of gravity**	**Status of G**	**Unification**
Newton / Einstein	Mass / stress-energy	Fundamental constant	None
Verlinde (entropic) [4]	Information entropy	Emergent (qualitative)	Qualitative
Jacobson (thermodynamic) [3]	Horizon thermodynamics	Proportionality	Qualitative
Padmanabhan [5]	Horizon / thermodynamics	Emergent (review)	Qualitative
This work	Harmonic closure volume	Derived: $\alpha\kappa\omega_m^2$	Quantitative

Table 5: Comparison with other emergent-gravity frameworks. "Quantitative" refers to internal program consistency, not independent experimental confirmation.

The most important comparison is with Verlinde's entropic gravity [4]. Verlinde derives an entropic force $F = T\,\Delta S/\Delta x$ that recovers Newton's second law near a holographic screen, where T is the Unruh temperature and ΔS is the change in entropy associated with displacing a test mass. The approach is elegant and physically motivated, but the temperature and entropy are assigned to macroscopic holographic screens without a microscopic derivation of why G takes its observed numerical value: G remains absorbed into the Bekenstein–Hawking entropy formula as a prior. Jacobson's derivation [3] faces the same issue: the Einstein equation emerges from $\delta Q = T\,\delta S$ applied to local Rindler horizons, but G enters through the identification of entropy with horizon area, which is itself proportional to $1/G$.

In contrast, the present framework assigns a specific geometric volume κ to each nucleon's closure state, making $G = \alpha\kappa\omega_m^2$ explicitly dependent on hadronic-scale physics. The value of G is then in principle sensitive to composition — a prediction absent from all thermodynamic approaches. The cost is the calibration of α, which replaces one unknown (G) with another (α). The gain is a concrete microscopic picture and a new falsifiable signature.

6.2 Physical Interpretation of $\alpha \sim 10^{-41}$

The extremely small value $\alpha \approx 2.35 \times 10^{-41}$ reflects the enormous hierarchy between microscopic confinement physics and macroscopic gravitational interaction. Within this relational interpretation, gravity is weak not because spacetime curvature is intrinsically small, but because only a tiny fraction of confinement information projects into the macroscopic relational sector.

[SPECULATIVE] More concretely: α measures the ratio of the gravitational energy scale to the hadronic confinement energy scale. The quantity $\kappa\omega_m^2$ carries dimensions of G and is of order 10^{30} $\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$ — some 41 orders of magnitude above G itself. Gravitational attraction therefore represents the large-scale tendency of relational systems to minimize informational differences between projected states, with α encoding how "inefficiently" that minimization couples to observable macroscopic acceleration.

[CALIBRATION] Whether this interpretation of α can be made rigorous — or whether α remains a brute calibration constant — is the model's most critical open question.

6.3 Relation to the Hierarchy Problem

[SPECULATIVE] The hierarchy between $\alpha \sim 10^{-41}$ and unity is reminiscent of the gauge hierarchy problem in particle physics (why $G_F \gg G_N$). This framework does not solve the hierarchy problem — it reframes it as a question about the projection efficiency of confinement information into macroscopic relational structure. Whether this reframing is more than a relabeling remains to be shown.

6.4 Cosmological Consistency Check

[SPECULATIVE] Using $M_U \sim 10^{53}$ kg (order-of-magnitude consistent with standard cosmological parameters [7]) and calibrated α, the closure depth estimate is $n \sim 10^{13} \approx 2^{45}$. This is an assumption-dependent plausibility estimate, not a standalone falsifiable prediction. Presented for transparency only.

6.5 Limitations and Future Work

Limitations:

- Weak-field, stationary regime only.
- α is calibrated, not derived (the model's single free parameter).
- Relativistic completion (coupling to a projection tensor $P_{\mu\nu}$) remains future work.

- Physical correspondence between the relational-charge ansatz and quantum field theory is not established.
- The n-selection rule is structurally motivated but not proven.
- Coefficients β, γ in Eq. (12) are not yet derived.

Future work:

- Derive α from a higher-order harmonic closure condition.
- Develop a strong-field tensor formulation for black holes and neutron stars.
- Quantitatively fit galaxy rotation curves using the $\kappa(r)$ ansatz.
- Establish the n-selection theorem.
- Map nuclear binding energy to ω_m variation for a quantitative WEP prediction.

7 Conclusion

We have presented a dimensionally consistent derivation of Newtonian gravity from relational projection principles. The key elements are:

1. **[DERIVED]** Relational charge density $\rho_{\text{rel}} = \kappa\omega_m^2\rho_m$, with $\kappa = v_{\text{id}}/m_p$.
2. **[DERIVED]** Poisson equation $\nabla^2\phi = 4\pi\alpha\kappa\omega_m^2\rho_m$ yielding Newtonian limit with $G = \alpha\kappa\omega_m^2$.
3. **[DERIVED]** κ determined geometrically from binary depth n: $\kappa = \frac{4\pi}{3}(2^{n/2}\bar{\lambda}_p)^3/m_p$.
4. **[DERIVED]** Identity radius $R_{\text{id}} = (3\kappa M/4\pi)^{1/3}$ emerging naturally from the closure-volume construction.
5. **[HYPOTHESIS]** Falsifiable avenues: composition-dependent WEP deviations, near-closure-scale corrections in neutron-star observations, and dark-matter-like rotation curves distinguishable from MOND.
6. **[CALIBRATION]** Quadripartite unification across 46 orders of magnitude from a single harmonic partition, with α as the sole calibrated parameter.

   "Gravity, like the size of the proton, the stability of atoms, and the expansion of the cosmos, is not a fundamental interaction but a manifestation of relational geometry seeking harmonic closure."

LLM Disclosure Statement

Models used: OpenAI ChatGPT, Anthropic Claude, Google Gemini. Sessions used for adversarial validation were conducted logged-out with blank context.

How they were employed:

- Dimensional-analysis consistency checking across all equations.
- Algebraic stress-testing: presenting each postulate chain and requesting identification of unjustified steps or circular reasoning.
- Literature synthesis: identifying prior work in emergent gravity (Verlinde, Jacobson, Padmanabhan) and equivalence-principle experiments (MICROSCOPE).
- **Adversarial Cross-Validation (AACV):** a separate session (logged out, blank context) was prompted with a neutral review task using the contest rubric criteria. Outputs were used to identify: the missing LLM disclosure in v1.0, the need for explicit α parameter accounting, and the speculative status of the dark-matter section.
- Language revision and LaTeX formatting.

Estimated contribution: $\sim 35\%$ drafting/structure; $\sim 65\%$ human-directed conceptual development, postulate formulation, and iterative refinement.

Human oversight: All mathematical claims were verified by the author independently. LLM outputs were treated as drafting aids and consistency probes, not epistemic authorities. Citation details were cross-checked against primary sources.

Community acknowledgment: Feedback from r/LLMPhysics contributed to structural improvements across earlier versions. This submission incorporates that feedback.

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